| Question | Answer | Marks | Partial Marks | Guidance |
|---|---|---|---|---|
| 2(a) | –4, –1 | **2** | **B1** for each correct value | |
| 2(b) | Correct graph | **4** | **B3FT** for 7 or 8 correct points<br>or **B2FT** for 5 or 6 correct points<br>or **B1FT** for 3 or 4 correct points | Mark curve first using overlay as a guide, accuracy 1mm radially.<br>Condone slight feathering in no more than 2 places.<br>No excessively thick curve or incorrect curvature.<br>No ruled sections<br>If not 4 marks, mark plots using same tolerance as the curve. If plot is large, look at centre of plot.<br>Curve can imply any missing plots.<br>No more than 1 cm of vertical section at ends<br>Ignore other curves/lines - could be part (d)<br>See AG |
| 2(c)(i) | Ruled tangent at $x = 1$ | **1** | | No gap at $x = 1$.<br>Mark not available if ruled line segments on graph between $x = 0$ and $x = 1.5$<br>See AG |
| 2(c)(ii) | 6 to 14 nfww | **2** | **dep** on correct tangent or a close attempt at the tangent at $x = 1$<br><br>**M1** for rise/run for *their* tangent, or close attempt at tangent at any point.<br>Must see correct or implied calculation from a drawn tangent. | If 1 mark awarded for tangent or a close attempt at tangent and gradient in range, award 2 marks <u>provided</u> no evidence of calculus **used or method other than** difference in $y$ / difference in $x$<br>If answer in range then no need to check points used in difference in $y$ / difference in $x$ calculation<br>Allow integer/integer if in range.<br><br>For M1 if no working shown, look at the value of their gradient first, then check one unit horizontally for their tangent. The absolute value of their gradient must be within 1 of your value. If no value then check working – must be correct.<br>See AG |

| Question | Answer | Marks | Partial Marks | Guidance |
|---|---|---|---|---|
| 2(d) | $y = x + 2$ ruled | **M2** | **M1** for [$y$ =] $x + 2$ soi<br><br>or $y = x + k$ ruled<br><br>or $y = kx + 2$ ruled, but not $y = 2$ | For M2 accuracy 1mm radially at (–2, 0) and at (0, 2) and must intersect curve at all possible places.<br><br>For soi allow written, freehand line, short line, or dashed line,<br>e.g. for M1 $x^3 + 4x^2 - 4 = x + 2$<br><br>Accuracy for M1: 1mm radially for *their* intercepts |
| | $x = -3.95$ to $-3.75$<br>$x = -1.4$ to $-1.25$<br>$x = 1.1$ to $1.25$ | **A2** | **A1** for any two values<br><br>If A0, **SC1** for three correct values | A1 dependent on M2<br><br>M1SC1 possible |